

ST PAUL ISLAND, NS, Canada

WMO: 714180

Lat: 47.226N

Lon: 60.140W

Elev: 89

StdP: 14.65

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.8	10.1	0.0	5.5	7.1	3.7	6.7	10.3	60.7	22.9	55.9	21.1	30.0	280	1.039

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	72.5	66.3	70.5	65.3	68.9	64.4	67.8	70.9	66.5	69.4	65.2	68.2	16.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.0	69.8	65.3	93.7	68.6	63.9	89.2	67.4	32.3	71.1	31.3	69.6	30.2	68.2	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
52.9	47.3	42.3	DB	3.9	77.5	6.1	2.1	-0.5	79.0	-4.0	80.3	-7.5	81.4	-11.9	82.9
			WB	3.0	70.1	6.7	1.0	-1.9	70.8	-5.8	71.3	-9.5	71.9	-14.4	72.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	25.9	23.3	27.2	33.8	41.5	50.7	61.6	64.7	58.2	49.1	40.2	32.2
	DBStd	14.84	7.03	7.39	6.42	4.54	4.67	5.68	4.71	3.76	4.59	5.08	5.73	5.84
	HDD50	3944	746	747	707	486	270	57	0	0	2	78	298	553
	HDD65	8284	1211	1167	1172	936	729	430	125	51	209	492	744	1018
	CDD50	1201	0	0	0	0	5	78	360	456	247	51	4	0
	CDD65	67	0	0	0	0	0	1	20	42	4	0	0	0
	CDH74	25	0	0	0	0	0	0	9	16	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	21.5	28.2	24.4	22.8	20.3	18.0	17.4	16.0	16.9	20.1	22.6	24.4	27.2
Precipitation	PrecAvg	59.1	5.8	5.2	4.6	4.4	4.0	4.1	3.5	3.9	4.7	6.0	6.0	6.8
	PrecMax	72.9	8.9	7.2	8.7	7.4	7.0	6.8	6.7	9.0	12.1	10.5	9.5	14.0
	PrecMin	45.3	2.8	1.9	1.9	2.3	1.9	2.0	1.5	1.7	1.7	2.4	3.8	3.8
	PrecStd	6.0	1.5	1.5	1.7	1.3	1.4	1.5	1.2	1.3	2.4	2.2	1.6	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	43.0	44.5	53.2	60.9	70.1	75.0	76.1	71.5	65.1	58.7	50.9
		MCWB	46.8	42.1	42.4	49.7	53.5	62.8	66.4	67.5	65.7	60.6	55.4	49.2
	2%	DB	41.2	37.4	39.3	45.6	54.7	65.3	71.9	73.1	68.3	60.6	53.3	44.7
		MCWB	40.2	36.2	37.8	42.4	50.2	59.0	65.6	66.7	64.1	57.0	51.3	43.5
	5%	DB	37.2	34.2	35.9	41.5	50.6	61.8	69.9	71.3	66.3	58.0	50.3	42.0
		MCWB	36.0	33.3	34.4	39.0	46.6	57.0	64.5	66.2	62.5	55.0	48.7	40.9
	10%	DB	34.5	32.0	33.7	38.6	47.4	58.5	68.0	69.7	64.2	55.8	47.7	39.9
		MCWB	33.5	31.2	32.5	36.7	43.9	54.4	63.7	65.2	60.6	53.0	46.2	38.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	42.1	43.4	50.7	55.0	65.0	69.2	69.8	67.3	61.8	55.8	49.4
		MCDB	49.1	42.7	45.5	52.4	57.5	68.7	72.8	73.2	70.2	64.3	57.9	50.9
	2%	WB	40.7	36.7	38.2	43.4	49.8	60.8	67.5	68.3	65.0	57.7	51.5	43.8
		MCDB	41.2	37.3	39.2	44.9	52.1	63.2	70.6	71.3	67.8	59.9	52.8	44.6
	5%	WB	36.5	33.6	34.8	39.5	46.4	57.3	65.9	67.1	63.3	55.5	48.9	41.2
		MCDB	37.2	34.2	35.7	41.1	49.0	60.6	68.8	70.0	65.8	57.5	50.0	42.0
	10%	WB	33.5	31.3	32.6	36.9	44.0	54.7	64.3	66.0	61.6	53.4	46.4	38.8
		MCDB	34.3	31.9	33.3	38.3	46.1	57.4	67.0	68.9	63.8	55.4	47.5	39.7
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.5	7.5	6.7	7.7	8.1	7.8	6.3	6.5	5.9	6.8	6.8
		MCDBR	14.9	13.4	11.7	13.2	14.1	14.2	10.7	8.6	9.2	9.5	11.9	11.8
		MCWBR	15.4	13.8	11.4	10.8	10.3	9.3	7.0	6.1	7.8	8.9	12.0	12.5
	5% WB	MCDBR	15.6	13.7	12.1	12.9	12.8	12.9	10.2	7.6	8.7	9.4	11.6	12.0
		MCWBR	15.7	13.9	11.6	11.3	10.5	9.9	7.6	6.4	7.9	9.1	11.9	12.6
Clear-Sky Solar Irradiance	taub	0.312	0.302	0.335	0.356	0.369	0.388	0.415	0.395	0.368	0.353	0.326	0.312	
	taud	2.378	2.401	2.320	2.402	2.444	2.399	2.372	2.433	2.480	2.455	2.452	2.376	
	Ebn at Noon	243	275	281	285	284	278	268	270	268	252	237	227	
	Edh at Noon	23	27	34	35	35	36	37	33	29	26	21	20	
All-Sky Solar Radiation	RadAvg	279	536	976	1351	1642	1728	1701	1526	1171	716	353	215	
	RadStd	24	60	93	109	136	173	131	102	88	59	41	20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	(e) +2.45	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +112	(m) +22
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page